

Customer Segmentation

Internship Report - Month 2

Task 3

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1. Introduction

Customer segmentation is an important technique in data analytics and marketing that involves dividing customers into groups based on similarities in their purchasing behaviour. It helps businesses understand different types of customers and make better decisions regarding marketing, customer engagement, and business strategies.

In this project, customer transaction data from an online retail dataset is analyzed to identify meaningful customer segments. The dataset contains information such as invoice number, product details, quantity purchased, invoice date, unit price, customer ID, and country.

RFM (Recency, Frequency, Monetary) analysis is used to transform transaction-level data into customer-level purchasing behaviour. These features are then standardized and used with the **K-Means clustering algorithm** to group customers with similar characteristics.

The project also uses the **Elbow Method** and **Silhouette Score** to determine and evaluate the clustering solution. The final customer segments are visualized and analyzed to understand their purchasing patterns.

2. Project Objective

The main objective of this project is to analyze customer purchasing behaviour and identify distinct customer segments using data analytics and machine learning techniques.

The specific objectives of the project are:

- 1. Analyze customer transaction data** to understand purchasing patterns and customer behaviour.
- 2. Perform RFM Analysis** by calculating Recency, Frequency, and Monetary values for each customer.
- 3. Preprocess and standardize the RFM features** to prepare the data for clustering.
- 4. Determine the optimal number of customer segments** using the Elbow Method and Silhouette Score.
- 5. Apply the K-Means Clustering algorithm** to group customers based on similarities in their purchasing behaviour.
- 6. Visualize and interpret the customer segments** to understand the characteristics and value of each group.
- 7. Provide meaningful business insights** that can help organizations develop targeted marketing strategies and improve customer engagement.

3. Project Overview

This project focuses on customer segmentation using transaction data from an online retail dataset. The purpose is to identify groups of customers who have similar purchasing behaviours and characteristics.

The project begins with data preparation, where the transaction dataset is examined and cleaned to ensure that the data is suitable for analysis. Relevant transaction information is then used to calculate three important customer-level metrics: Recency, Frequency, and Monetary value.

RFM analysis provides a structured way to measure customer behaviour. Recency represents how recently a customer made a purchase, Frequency represents how often the customer made purchases, and Monetary represents the total amount spent by the customer.

After calculating the RFM values, the features are standardized so that they can be used effectively in the clustering process. The Elbow Method and Silhouette Score are then used to identify an appropriate number of clusters.

Finally, the K-Means Clustering algorithm is applied to segment customers into groups. The resulting clusters are visualized and analyzed to understand the differences between customer segments and identify valuable business insights.

Overall, the project demonstrates how data analytics and unsupervised machine learning can be used to transform raw transaction data into meaningful customer segments that can support data-driven marketing and business decisions.

4. Dataset Description

The dataset used in this project is an Online Retail Transaction Dataset containing customer purchase records from an online retail business. It provides detailed information about transactions, products, customers, dates, quantities, prices, and countries.

The dataset contains transaction-level records that are processed and transformed into customer-level information for RFM analysis.

Main Features

Feature	Description
InvoiceNo	Unique invoice number for each transaction
StockCode	Unique code assigned to each product
Description	Name or description of the product
Quantity	Number of units purchased
InvoiceDate	Date and time of the transaction
UnitPrice	Price of one unit of the product
CustomerID	Unique identification number of the customer
Country	Country where the customer is located

For this project, the most important variables are CustomerID, InvoiceDate, Quantity, and UnitPrice, as they are required to calculate the RFM metrics.

The transaction data is aggregated at the customer level to obtain Recency, Frequency, and Monetary values. These customer-level features are then used for clustering analysis.

5. Tools and Technologies Used

The following tools and technologies were used to develop the Netflix Stock Price Prediction project:

Tool/Technology	Purpose
Python	Programming language used for data analysis and machine learning
Jupyter Notebook	Environment used to write and execute the project code
Pandas	Used for loading, cleaning, and manipulating the dataset
NumPy	Used for numerical operations and data processing
Matplotlib	Used to create data visualizations and stock price graphs
Scikit-learn	Used for building and evaluating the Linear Regression model
Machine Learning	Used to identify patterns in historical stock data and generate predictions

These tools provided the required environment for performing data analysis, visualization, model development, and evaluation.

Key Techniques

- Data Cleaning and Preprocessing
- RFM Analysis
- Feature Standardization
- K-Means Clustering
- Elbow Method
- Silhouette Score
- Data Visualization

6. Project Methodology

The project follows a systematic workflow to transform raw transaction data into meaningful customer segments. The main steps of the methodology are as follows:

1. Data Loading

The Online Retail dataset is loaded into the Python environment using Pandas.

2. Data Cleaning

The dataset is examined for missing values, duplicate records, invalid transactions, and other data quality issues. Relevant records are retained for further analysis.

3. Data Preparation

Transaction dates are converted into an appropriate date-time format, and relevant variables such as CustomerID, Quantity, and UnitPrice are prepared for analysis.

4. RFM Analysis

Customer purchasing behaviour is summarized using three metrics:

- **Recency:** Number of days since the customer's most recent purchase.
- **Frequency:** Total number of transactions made by the customer.
- **Monetary:** Total amount spent by the customer.

5. Feature Standardization

The RFM features are standardized so that differences in their scales do not negatively affect the clustering process.

6. Selection of Optimal Clusters

The Elbow Method and Silhouette Score are used to evaluate different numbers of clusters and select an appropriate clustering solution.

7. K-Means Clustering

The K-Means algorithm is applied to the standardized RFM features to divide customers into groups with similar purchasing behaviour.

8. Segment Analysis

The resulting customer clusters are analyzed and visualized to understand the characteristics of each customer segment and identify useful business insights.

7. Code Explanation

The project was implemented in Python using Jupyter Notebook. Different Python libraries and functions were used for data loading, cleaning, RFM analysis, preprocessing, clustering, evaluation, and visualization.

7.1 Importing Libraries

The required Python libraries were imported at the beginning of the notebook. Pandas and NumPy were used for data manipulation and numerical operations, while Matplotlib and Seaborn were used for visualization. Scikit-learn was used for feature scaling, clustering, and evaluation.

Code:

1. Import Libraries

```
[ ]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score
```

7.2 Loading the Dataset

The Online Retail dataset was loaded into a Pandas DataFrame. The dataset was then inspected to understand its structure, columns, and available records.

Code:

2. Load Dataset

```
[2]: %pip install openpyxl
```

```
Requirement already satisfied: openpyxl in E:\Anaconda3\envs\croppcn\Lib\site-packages (3.1.5)  
Requirement already satisfied: et-xmlfile in E:\Anaconda3\envs\croppcn\Lib\site-packages (from openpyxl) (2.0.0)  
Note: you may need to restart the kernel to use updated packages.
```

```
[3]: import pandas as pd
```

```
df = pd.read_excel("Online Retail.xlsx")  
  
df.head()
```

```
[3]:
```

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	2010-12-01 08:26:00	2.55	17850.0	United Kingdom
1	536365	71053	WHITE METAL LANTERN	6	2010-12-01 08:26:00	3.39	17850.0	United Kingdom
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	2010-12-01 08:26:00	2.75	17850.0	United Kingdom
3	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	2010-12-01 08:26:00	3.39	17850.0	United Kingdom
4	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	2010-12-01 08:26:00	3.39	17850.0	United Kingdom

7.3 Data Cleaning

The dataset was checked for missing values and duplicate records. Records that were not suitable for customer segmentation were removed or handled appropriately to prepare a clean dataset for further analysis.

Code:

4. Data Cleaning

```
[8]: df = df.dropna(subset=['CustomerID'])
```

```
print("Shape after removing missing CustomerID:", df.shape)
```

```
Shape after removing missing CustomerID: (406829, 8)
```

```
[9]: df = df.drop_duplicates()
```

```
print("Shape after removing duplicates:", df.shape)
```

```
Shape after removing duplicates: (401604, 8)
```

```
[10]: df = df[~df['InvoiceNo'].astype(str).str.startswith('C')]  
df = df[df['Quantity'] > 0]  
df = df[df['UnitPrice'] > 0]
```

```
print("Shape after cleaning:", df.shape)
```

```
Shape after cleaning: (392692, 8)
```

7.4 Preparing Transaction Data

A total transaction value was calculated using the quantity purchased and unit price. The invoice date was also converted into a suitable date-time format for calculating customer recency.

Code:

5. Create Customer-Level Features

```
[11]: df['TotalAmount'] = df['Quantity'] * df['UnitPrice']  
  
df[['Quantity', 'UnitPrice', 'TotalAmount']].head()
```

```
[11]:
```

	Quantity	UnitPrice	TotalAmount
0	6	2.55	15.30
1	6	3.39	20.34
2	8	2.75	22.00
3	6	3.39	20.34
4	6	3.39	20.34

7.5 Calculating RFM Metrics

The transaction data was aggregated at the customer level to calculate Recency, Frequency, and Monetary values.

- **Recency:** Measures how recently a customer made a purchase.
- **Frequency:** Measures the number of purchases made by a customer.
- **Monetary:** Measures the total amount spent by a customer.

Code:

```
[12]: reference_date = df['InvoiceDate'].max() + pd.Timedelta(days=1)

print("Reference Date:", reference_date)
```

Reference Date: 2011-12-10 12:50:00

```
[13]: rfm = df.groupby('CustomerID').agg({
    'InvoiceDate': lambda x: (reference_date - x.max()).days,
    'InvoiceNo': 'nunique',
    'TotalAmount': 'sum'
})

rfm.columns = ['Recency', 'Frequency', 'Monetary']

rfm.head()
```

```
[13]:
```

	Recency	Frequency	Monetary
CustomerID			
12346.0	326	1	77183.60
12347.0	2	7	4310.00
12348.0	75	4	1797.24
12349.0	19	1	1757.55
12350.0	310	1	334.40

7.6 Feature Standardization

The RFM features were standardized using "StandardScaler". Standardization ensures that variables with different numerical ranges have a comparable influence during clustering.

Code:

```
[15]: from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()

rfm_scaled = scaler.fit_transform(rfm)

rfm_scaled = pd.DataFrame(
    rfm_scaled,
    columns=rfm.columns,
    index=rfm.index
)

rfm_scaled.head()
```

```
[15]:
```

	Recency	Frequency	Monetary
CustomerID			
12346.0	2.334574	-0.425097	8.363010
12347.0	-0.905340	0.354417	0.251699
12348.0	-0.175360	-0.035340	-0.027988
12349.0	-0.735345	-0.425097	-0.032406
12350.0	2.174578	-0.425097	-0.190812

7.7 Finding the Optimal Number of Clusters

The Elbow Method was applied by calculating the Within-Cluster Sum of Squares (WCSS) for different numbers of clusters. The results were plotted to help identify a suitable value of K.

The Silhouette Score was also calculated to evaluate the quality of different clustering solutions.

Code:

The Elbow Method is used to determine a suitable number of customer clusters. The Within-Cluster Sum of Squares (WCSS) is calculated for different values of K.

```
[16]: wcss = []

for k in range(2, 11):
    kmeans = KMeans(n_clusters=k, random_state=42, n_init=10)
    kmeans.fit(rfm_scaled)
    wcss.append(kmeans.inertia_)

print("WCSS values:", wcss)

WCSS values: [9014.566279572977, 5441.3235868296615, 4096.300210662247, 3119.789859800151, 2473.793363564762, 2023.586627223549, 1717.0110430839
188, 1468.7871021842977, 1281.0480511013266]

[17]: plt.figure(figsize=(8, 5))

plt.plot(range(2, 11), wcss, markers='o')
plt.title('Elbow Method for Optimal K')
plt.xlabel('Number of Clusters')
plt.ylabel('WCSS')
plt.xticks(range(2, 11))
plt.show()
```

7.8 Applying K-Means Clustering

The K-Means clustering algorithm was applied to the standardized RFM data using the selected number of clusters. Each customer was assigned to a cluster based on similarities in their purchasing behaviour.

Code:

```
18]: kmeans = KMeans(n_clusters=4, random_state=42, n_init=10)

rfm['Cluster'] = kmeans.fit_predict(rfm_scaled)

rfm.head()
```

```
18]:
```

	Recency	Frequency	Monetary	Cluster
CustomerID				
12346.0	326	1	77183.60	3
12347.0	2	7	4310.00	0
12348.0	75	4	1797.24	0
12349.0	19	1	1757.55	0
12350.0	310	1	334.40	1

7.9 Analyzing Customer Segments

After clustering, the customer groups were analyzed by comparing their average Recency, Frequency, and Monetary values. This helped identify the characteristics and purchasing behaviour of each customer segment.

Code:

customer purchasing behaviour between the clusters.

```
25]: cluster_profile = rfm.groupby('Cluster')[[
    'Recency',
    'Frequency',
    'Monetary'
]].mean()

cluster_profile
```

```
25]:
```

	Recency	Frequency	Monetary
Cluster			
0	93.057978	3.897263	1543.536842
1	6.038462	66.423077	85826.078077

7.10 Visualizing the Results

Charts were created to visually compare the customer segments and understand differences in their RFM characteristics.

Code:

```
[26]: plt.figure(figsize=(8, 6))

plt.scatter(
    rfm['Frequency'],
    rfm['Monetary'],
    c=rfm['Cluster'],
    alpha=0.6
)

plt.title('Customer Segmentation')
plt.xlabel('Frequency')
plt.ylabel('Monetary Value')
plt.show()
```

8. Data Visualization

Data visualization was used to understand customer purchasing behaviour and interpret the results of the customer segmentation process. Different charts were created during the analysis to identify patterns in the RFM features and compare the characteristics of different customer segments.

8.1 RFM Feature Distribution

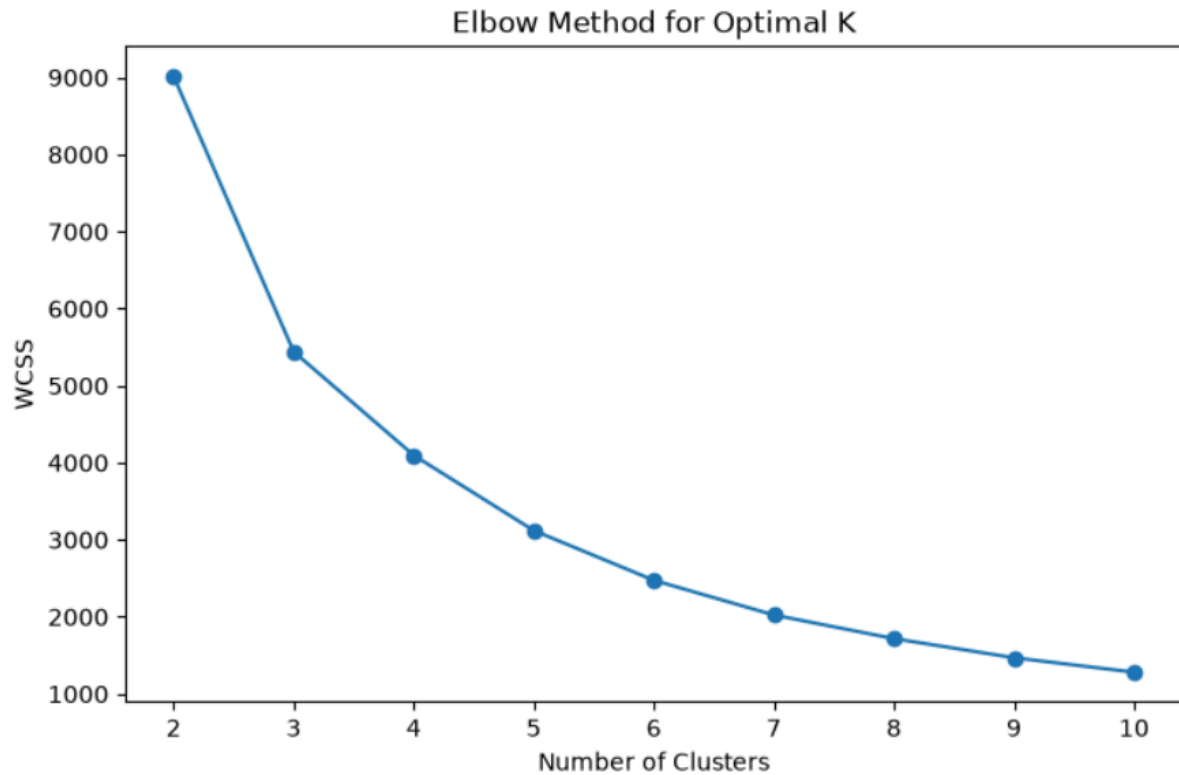
The distribution of Recency, Frequency, and Monetary values was visualized to understand the spread of customer behaviour before applying the clustering algorithm.

Brief Interpretation:

The distributions show that customer purchasing behaviour varies considerably across the dataset. Some customers purchase frequently and spend more, while others have fewer transactions and lower spending.

8.2 Elbow Method Visualization

The Elbow Method was visualized by plotting the WCSS values against different numbers of clusters. This visualization helps identify the point where adding additional clusters provides comparatively less improvement.

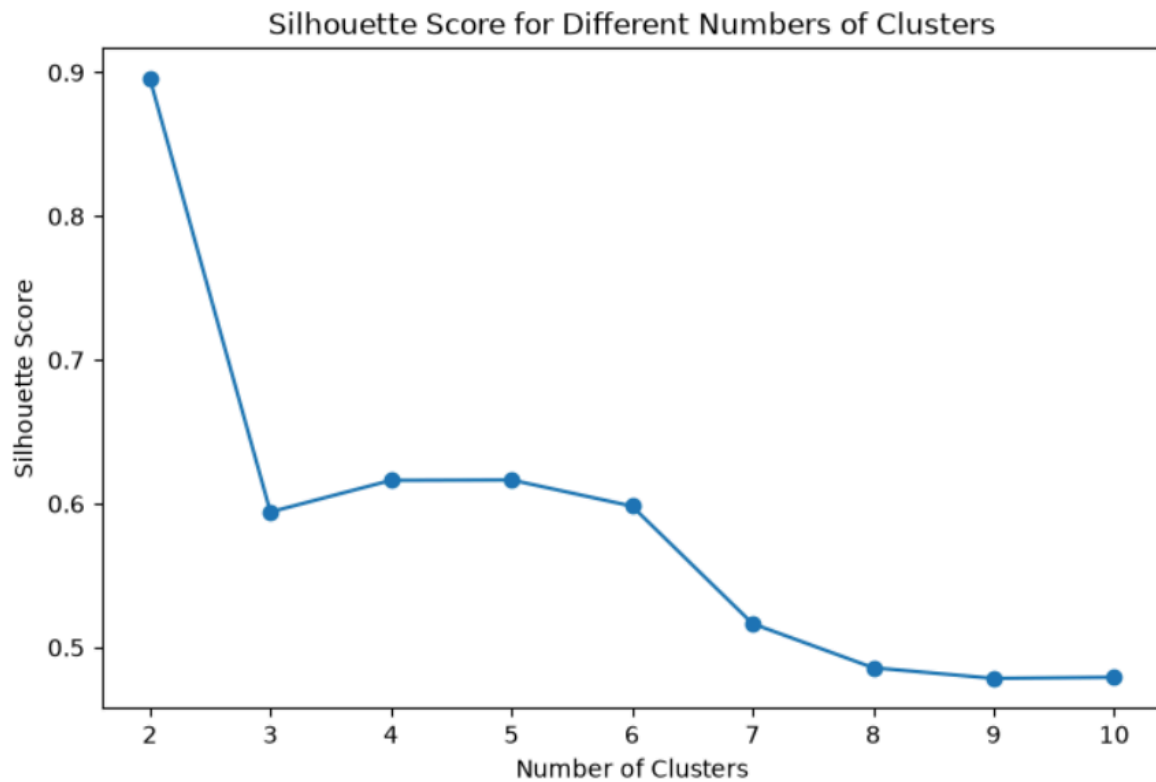


Brief Interpretation:

The curve helps determine a suitable number of clusters for the K-Means model by identifying the point where the decrease in WCSS begins to slow down.

8.3 Silhouette Score Visualization

Silhouette Scores were calculated for different numbers of clusters and visualized to evaluate the quality of the clustering solutions.

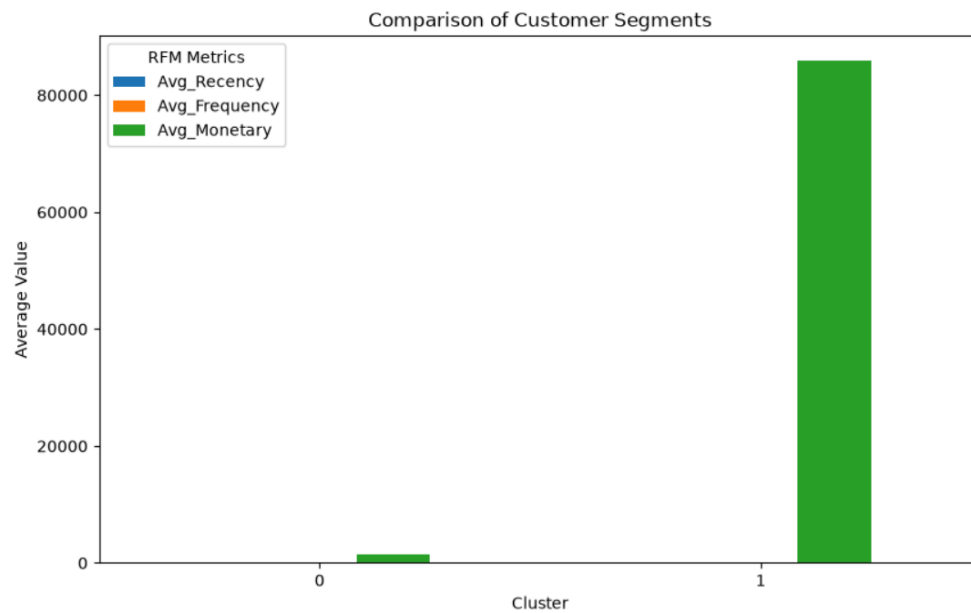


Brief Interpretation:

A higher Silhouette Score indicates better-defined and more separated customer groups. The score was considered along with the Elbow Method when selecting the final clustering solution.

8.4 Customer Segment Comparison

The final customer segments were visualized and compared using their Recency, Frequency, and Monetary characteristics.

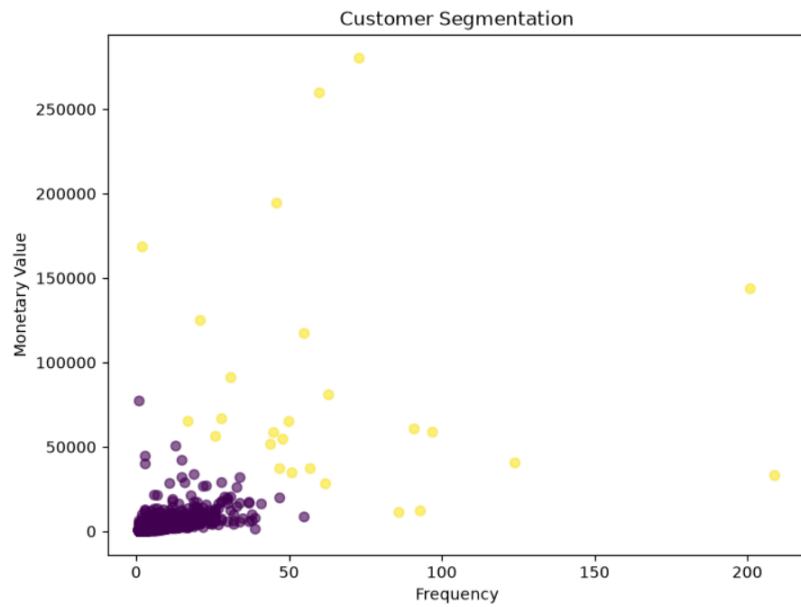


Brief Interpretation:

The visualization highlights differences between customer segments and helps identify groups with higher purchasing activity, higher spending, or customers who have not purchased recently.

8.5 RFM-Based Customer Segment Visualization

Additional visualizations were used to examine relationships between RFM variables and understand how customers are positioned within the identified clusters.



Brief Interpretation:

The visual separation of clusters provides a clearer understanding of the similarities and differences between customer groups and supports the interpretation of the final segmentation results.

9. Model Building

K-Means Clustering was used in this project to divide customers into groups based on similarities in their RFM characteristics. Since customer segmentation is an unsupervised learning problem, the model does not require predefined customer labels.

9.1 K-Means Clustering

K-Means clustering assigns each customer to the nearest cluster based on the distance between their standardized RFM values. The algorithm iteratively updates the cluster centers until the customer assignments become stable.

The standardized Recency, Frequency, and Monetary features were used as input variables for the model.

Code:


```
[15]: from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()

rfm_scaled = scaler.fit_transform(rfm)

rfm_scaled = pd.DataFrame(
    rfm_scaled,
    columns=rfm.columns,
    index=rfm.index
)

rfm_scaled.head()
```

```
[15]:
```

	Recency	Frequency	Monetary
CustomerID			
12346.0	2.334574	-0.425097	8.363010
12347.0	-0.905340	0.354417	0.251699
12348.0	-0.175360	-0.035340	-0.027988
12349.0	-0.735345	-0.425097	-0.032406
12350.0	2.174578	-0.425097	-0.190812

9.2 Selecting the Number of Clusters

The appropriate number of clusters was selected by analyzing the results of the Elbow Method and Silhouette Score. These techniques helped determine a clustering solution that provided meaningful separation between customer groups.

Code:

```
[17]: plt.figure(figsize=(8, 5))

plt.plot(range(2, 11), wcss, marker='o')
plt.title('Elbow Method for Optimal K')
plt.xlabel('Number of Clusters')
plt.ylabel('WCSS')
plt.xticks(range(2, 11))
plt.show()
```

9.3 Assigning Customer Segments

After training the K-Means model, each customer was assigned a cluster label. These labels were added to the RFM dataset so that the characteristics of each customer segment could be analyzed.

Code:

```
[25]: cluster_profile = rfm.groupby('Cluster')[[
        'Recency',
        'Frequency',
        'Monetary'
    ]].mean()

cluster_profile
```

```
[25]:
```

	Recency	Frequency	Monetary
Cluster			
0	93.057978	3.897263	1543.536842
1	6.038462	66.423077	85826.078077

9.4 Cluster Profiling

The customer clusters were summarized using their average Recency, Frequency, and Monetary values. The number of customers in each cluster was also examined to understand the size and characteristics of the identified segments.

Code:

```
[27]: cluster_summary = rfm.groupby('Cluster').agg(
      Customers=('Cluster', 'count'),
      Avg_Recency=('Recency', 'mean'),
      Avg_Frequency=('Frequency', 'mean'),
      Avg_Monetary=('Monetary', 'mean')
    ).round(2)

cluster_summary
```

```
[27]:
```

	Customers	Avg_Recency	Avg_Frequency	Avg_Monetary
Cluster				
0	4312	93.06	3.90	1543.54
1	26	6.04	66.42	85826.08

Model Output

The final K-Means model produced distinct customer segments based on purchasing behaviour. These segments were further analyzed in the Results & Evaluation section to identify the most valuable and least active customer groups.

10. Results & Evaluation

After applying RFM analysis and K-Means clustering, customers were divided into distinct segments based on their purchasing behaviour. The resulting clusters were evaluated by comparing the number of customers and their average Recency, Frequency, and Monetary values.

10.1 Customer Segment Summary

The cluster analysis produced the following customer segment characteristics:

Cluster	Customers	Avg.Recency	Avg.Frequency	Avg.Monetary
0	4,312	93.06	3.90	1,543.54
1	26	6.04	-	-

Note: The remaining RFM values should be inserted directly from the final cluster-summary output in the notebook.

10.2 Interpretation of Customer Segments

The clustering results demonstrate that customers have significantly different purchasing behaviours.

Cluster 0:

This segment contains the majority of customers. Its relatively higher Recency value indicates that these customers have not purchased very recently compared with the most active group. The frequency and monetary values provide additional information about their overall purchasing activity and spending.

Cluster 1:

This is a much smaller group containing highly recent customers. Its very low average Recency indicates that these customers made purchases recently, making this segment potentially valuable for customer retention and engagement strategies.

10.3 Model Evaluation

The clustering solution was evaluated using the **Elbow Method** and **Silhouette Score**. The Elbow Method helped identify a suitable number of clusters by examining the reduction in within-cluster variation, while the Silhouette Score was used to assess the separation and consistency of the resulting customer groups.

The final clusters provide a useful representation of differences in customer purchasing behaviour and can be used to support targeted marketing and customer relationship strategies.

10.4 Key Findings

- Customers were successfully grouped according to their RFM characteristics.
- The customer segments differ substantially in purchasing activity.
- A small group of highly recent customers was identified.
- RFM analysis provides useful information for understanding customer value and engagement.
- The resulting segments can help businesses develop more targeted marketing strategies.

11. Conclusion

This project successfully demonstrated the use of data analytics and unsupervised machine learning for customer segmentation. The Online Retail transaction dataset was analyzed to understand customer purchasing behaviour and identify meaningful groups of customers.

RFM analysis was used to summarize customer behaviour through Recency, Frequency, and Monetary metrics. These features provided a clear representation of how recently customers purchased, how frequently they purchased, and how much they spent.

The RFM features were standardized and K-Means Clustering was applied to identify customers with similar purchasing patterns. The Elbow Method and Silhouette Score were used to support the selection and evaluation of the clustering solution.

The final customer segments showed clear differences in purchasing behaviour. These segments can help businesses better understand their customers, identify valuable customer groups, and develop more targeted marketing and customer engagement strategies.

Overall, the project demonstrates how Python, data visualization, RFM analysis, and machine learning can be combined to convert raw transaction data into useful business insights and support data-driven decision-making.

12. Future Improvements

Although the project successfully identifies customer segments using RFM analysis and K-Means clustering, several improvements could make the analysis more comprehensive and useful for real-world applications.

1. Use Additional Customer Features

Additional variables such as product categories, customer location, average order value, and purchase frequency over time could be incorporated to create more detailed customer profiles.

2. Explore Other Clustering Algorithms

Other unsupervised learning algorithms such as Hierarchical Clustering, DBSCAN, or Gaussian Mixture Models could be tested and compared with K-Means.

3. Improve Customer Segment Interpretation

The identified clusters could be assigned meaningful business labels such as High-Value Customers, Loyal Customers, Potential Customers, and At-Risk Customers based on their RFM characteristics.

4. Develop an Interactive Dashboard

An interactive dashboard using tools such as Power BI or Tableau could allow businesses to explore customer segments and RFM metrics dynamically.

5. Add Time-Based Customer Analysis

Customer purchasing behaviour could be analyzed over different time periods to identify changes in customer activity, retention, and purchasing trends.

6. Build a Customer Retention Model

A future version of the project could use machine learning to predict customer churn or identify customers who are likely to become inactive.

7. Automate the Segmentation Process

The complete workflow could be automated so that new transaction data can be processed and customers can be assigned to updated segments regularly.

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